Design and Analysis of Randomized Trials in Public Health

Homework 11

1. **Refer to our class example, "foodsecurity.sas" from Topic 9 (RCBDs).** Use the following code to recode AEZ and garden into a single variable, AEZgarden:

```
data hfias; set hfias;
       if AEZ="high" and garden=0 then AEZgarden="high0";
       if AEZ="high" and garden=1 then AEZgarden="high1";
       if AEZ="med" and garden=0 then AEZgarden="med0";
       if AEZ="med" and garden=1 then AEZgarden="med1";
       if AEZ="low" and garden=0 then AEZgarden="low0";
       if AEZ="low" and garden=1 then AEZgarden="low1";
run;
```

Reanalyze these data as a one-way ANOVA using AEZgarden, then compare the results to our earlier output. Use a Tukey adjustment when you do pairwise comparisons of means; suggested code:

```
lsmeans AEZgarden / stderr tdiff adjust=tukey lines;
```

a) How do the two ANOVA tables compare? (2 points)

b) In our class example, we had the following output from the "slice" option. What is the null hypothesis (in words) for the first statistical test reported below (where AEZ = "high")? (2 points)

c) In your new SAS output, show the output that corresponds to the statistical test in part (b). What is the test statistic being used in your new output? How does it relate to the test statistic in part (b)? (2 points)

2. Recall the cluster RCT example we discussed in class, "hfias cluster randomization.sas" (Topic 11).

   a) We analyzed these data using Proc MIXED. Reanalyze the data, still using Proc MIXED, but ignore the clustering and treat the study as a completely randomized design. Make sure to keep the line "`lsmeans intervention / cl;`". For each model, what is your estimate for σ^2? Which model as a higher estimate, and why? (3 points)

   b) Compare the least squares means for the intervention and control groups in the two models. Does ignoring the clustering change your estimated average food insecurity in each group? Does ignoring the clustering change the standard errors of those estimates? In what way? (2 points)

c) If you incorrectly calculate the standard error of an estimate to be smaller than it actually is, you are going to erroneously think you have more precise estimates than you actually do. What are the (statistical) risks of thinking you have more precision than you actually do? (2 points)

d) Compare the 95% confidence intervals for the group means in the two models. Which model gives you wider confidence intervals? Why? (2 points)

3. You have developed an intervention intended to increase the amount of time patients with chronic back pain do a particular type of exercise, and you evaluate your intervention compared to standard of care physical therapy in a crossover study. In the dataset "backpain", "person" is the participant ID, and you have 20 participants. Every participant receives both the intervention (treatment="A") and the standard of care (treatment="B") in randomized order. "Period" indicates the time period in which your outcome, Y=number of minutes per week spent doing the recommended exercise, is assessed. What effect does your intervention have on the amount of time patients do the recommended exercise? Detail your steps, reasoning, and interpretation of results, as well as things you explore and check along the way. Include the SAS code you used. (10 points)

Homework 11

Alexandra Chang
PUBH 526: Design and Analysis of Randomized Trials in Public Health

December 8, 2025

Refer to our class example, "foodsecurity.sas" from Topic 9 (RCBDs). Use the following code to recode AEZ and garden into a single variable, AEZgarden:

data hfias; set hfias;
if AEZ="high" and garden=0 then AEZgarden="high0";
if AEZ="high" and garden=1 then AEZgarden="high1";
if AEZ="med" and garden=0 then AEZgarden="med0";
if AEZ="med" and garden=1 then AEZgarden="med1";
if AEZ="low" and garden=0 then AEZgarden="low0";
if AEZ="low" and garden=1 then AEZgarden="low1";
run;

Reanalyze these data as a one-way ANOVA using AEZgarden, then compare the results to our earlier output. Use a Tukey adjustment when you do pairwise comparisons of means; suggested code: *lsmeans AEZgarden / stderr tdiff adjust=tukey lines;*

Question 1.a (2 points)

a) How do the two ANOVA tables compare?

The ANOVA table from the original RCBD model (including `AEZ`, `garden`, and their interaction) and the ANOVA table from the one-way ANOVA using the combined factor `AEZgarden` are numerically identical. In both analyses, the model sum of squares (839.24), error sum of squares (5471.72), total sum of squares (6310.96), degrees of freedom (5 and 594), mean square error (9.21), and overall F statistic ($F = 18.22$, $p < 0.0001$) are the same.

This equivalence occurs because `AEZgarden` encodes the six combinations of `AEZ` and `garden`. Fitting the model

$$\text{foodsec} = \text{AEZ} + \text{garden} + \text{AEZ} \times \text{garden}$$

is therefore algebraically equivalent to fitting the one-way model

foodsec = AEZgarden.

As a result, the two ANOVA tables match exactly.

Question 1.b (2 points)

In our class example, we had the following output from the "slice" option. What is the null hypothesis (in words) for the first statistical test reported below (where AEZ = "high")?

The first slice test at AEZ = "high" evaluates whether the garden intervention has any effect on food security within the high–AEZ stratum. Therefore, the null hypothesis is:

$$H_0 : \text{Among households in the high AEZ, the mean food security score is the same for garden = 0 and garden = 1.}$$

In words: *Within the high agroecological zone, there is no difference in mean food security between households with the garden intervention and those without it.*

Question 1.c (2 points)

In your new SAS output, show the output that corresponds to the statistical test in part (b). What is the test statistic being used in your new output? How does it relate to the test statistic in part (b)?

In the one-way ANOVA using `AEZgarden`, the hypothesis from part (b) (comparing garden = 0 vs. garden = 1 within the high AEZ) corresponds to the comparison between the levels `high0` and `high1` in the Tukey-adjusted pairwise LSMEANS table.

From the new SAS output, the relevant line is:

$$t = 6.288367, \qquad p < 0.0001$$

This t–statistic is numerically identical to the test statistic from the slice option in part (b). Both tests evaluate the same contrast: the difference in mean food security between garden = 0 and garden = 1 within the high AEZ. The one-way ANOVA expresses this contrast using the combined factor `AEZgarden`, but the underlying comparison and test statistic are the same. If you want, I can also generate a LaTeX table for the relevant row of the LSMEANS matrix or format your entire homework document.

Recall the cluster RCT example we discussed in class, "hfias cluster randomization.sas" (Topic 11).

Question 2.a (3 points)

We analyzed these data using `Proc MIXED`. Reanalyze the data, still using `Proc MIXED`, but ignore the clustering and treat the study as a completely randomized design. Make sure to keep the line `lsmeans intervention / cl;`. For each model, what is your estimate for σ^2? Which model has a higher estimate, and why?

From the clustered model fit using `Proc MIXED`, the covariance parameter table reports a residual variance estimate of

$$\hat{\sigma}^2_{\text{clustered}} = 9.1826.$$

When the model is refit as a completely randomized design (ignoring village-level clustering), the residual variance estimate increases to

$$\hat{\sigma}^2_{\text{CRD}} = 12.6445.$$

The completely randomized design yields the larger estimate of σ^2 because ignoring clustering forces all between-village variation to be absorbed into the residual error term. In contrast, the clustered model includes a random intercept for villages, which captures between-cluster variability (estimated as 3.8145) and therefore leaves less unexplained variability at the individual level. Thus,

$$\hat{\sigma}^2_{\text{CRD}} > \hat{\sigma}^2_{\text{clustered}}.$$

Question 2.b (2 points)

Compare the least squares means for the intervention and control groups in the two models. Does ignoring the clustering change your estimated average food insecurity in each group? Does ignoring the clustering change the standard errors of those estimates? In what way?

The clustered model and the completely randomized design (CRD) model produce the same least squares mean estimates for the intervention and control groups:

$$\widehat{\mu}_{\text{control}} = 11.79, \qquad \widehat{\mu}_{\text{intervention}} = 6.64.$$

Thus, ignoring the clustering does *not* change the point estimates of the group means.

However, ignoring clustering has a substantial effect on the standard errors. From the clustered model:

$$SE_{\text{clustered}} = 0.6767,$$

while the CRD model produces much smaller standard errors:

$$SE_{\text{CRD}} = 0.3246.$$

The standard errors decrease by more than half when clustering is ignored. This occurs because the CRD model treats all 240 observations as independent, failing to account for within-village correlation. In the clustered model, the random intercept absorbs between-village variability, leading to larger and more realistic standard errors.

In summary: the estimated group means stay the same, but the standard errors become artificially small when clustering is ignored, overstating the precision of the estimates.

Question 2.c (2 points)

If you incorrectly calculate the standard error of an estimate to be smaller than it actually is, you are going to erroneously think you have more precise estimates than you actually do. What are the (statistical) risks of thinking you have more precision than you actually do?

If the standard error is underestimated, the analysis will suggest that the estimate is more precise than it truly is. Statistically, this leads to several risks:

- **Inflated Type I error rate:** confidence intervals become too narrow and hypothesis tests are more likely to reject the null, even when the null is true.
- **Overconfident conclusions:** researchers may conclude that an intervention has a significant effect when the apparent significance is due only to the underestimated variability.
- **Misleading inference:** p-values become artificially small, exaggerating the strength of evidence against the null hypothesis.

In short, underestimating the standard error creates a false sense of precision and increases the chance of making incorrect statistical conclusions.

Question 2.d (2 points)

Compare the 95% confidence intervals for the group means in the two models. Which model gives you wider confidence intervals? Why?

From the clustered model, the 95% confidence intervals for the group means are:

$$\text{Control (no): } [10.37,\ 13.21], \qquad \text{Intervention (yes): } [5.22,\ 8.06].$$

From the completely randomized design (CRD) model that ignores clustering, the intervals are narrower:

Control (no): [11.15, 12.43], Intervention (yes): [6.00, 7.28].

The clustered model produces the *wider* confidence intervals. This is because the clustered model correctly accounts for correlation among households within the same village by including a random intercept. Accounting for this intra-cluster correlation increases the standard errors, which in turn widens the confidence intervals.

In contrast, the CRD model incorrectly treats all observations as independent, underestimates the standard errors, and therefore produces confidence intervals that are unrealistically narrow.

Question 3 (10 points)

You have developed an intervention intended to increase the amount of time patients with chronic back pain do a particular type of exercise, and you evaluate your intervention compared to standard of care physical therapy in a crossover study. In the dataset "backpain", "person" is the participant ID, and you have 20 participants. Every participant receives both the intervention (treatment="A") and the standard of care (treatment="B") in randomized order. "Period" indicates the time period in which your outcome, Y=number of minutes per week spent doing the recommended exercise, is assessed. What effect does your intervention have on the amount of time patients do the recommended exercise? Detail your steps, reasoning, and interpretation of results, as well as things you explore and check along the way. Include the SAS code you used.

SAS Code Used

```
libname mydata '/home/u64258127/sasuser.v94';

data backpain;
    set mydata.backpain;
run;

/* Descriptive summaries */
proc contents data=backpain; run;
proc means data=backpain; var Y; class treatment period; run;

/* Crossover analysis */
proc mixed data=backpain;
    class person treatment period;
    model Y = treatment period / ddfm=kr;
    random person;
    lsmeans treatment / cl;
run;
```

Model Interpretation

Preliminary summaries showed noticeable differences in mean exercise time across periods, suggesting that period effects should be included in the model. The mixed model estimates person-level heterogeneity as a random effect and adjusts the fixed treatment comparison accordingly.

The variance components were:

$$\hat{\sigma}^2_{\text{person}} = 8.5652, \qquad \hat{\sigma}^2_{\text{residual}} = 43.8222,$$

indicating meaningful between-person variability.

Estimated Treatment Effect

Least-squares means from the mixed model were:

$$\hat{\mu}_A = 44.3 \text{ minutes/week}, \qquad \hat{\mu}_B = 49.4 \text{ minutes/week}.$$

Thus, the estimated treatment difference was:

$$\hat{\mu}_A - \hat{\mu}_B = -5.1 \text{ minutes/week}.$$

The Type III test for treatment yielded:

$$F(1, 18) = 5.94, \quad p = 0.0255,$$

indicating a statistically significant difference between treatments after adjusting for period.

Period Effect

There was also a significant period effect:

$$F(1, 18) = 10.86, \quad p = 0.0040,$$

with participants exercising more in Period 2 regardless of treatment. This supports the decision to include period as a fixed effect in the model.

Conclusion

After accounting for both period effects and within-person correlation, the intervention (Treatment A) did not increase exercise time. In fact, participants performed approximately 5 fewer minutes of recommended exercise per week under Treatment A compared to standard care. The effect was statistically significant.

Conclusion: Treatment A resulted in lower weekly exercise time than Treatment B.

See SAS codes/outputs below:

```
/*************************************************************************

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    Health
 * Homework 11
 * Author:  Alexandra Chang
 * Dataset: hfias.sas7bdat / backpain.sas7bdat
 *************************************************************************/


libname mydata '/home/u64258127/sasuser.v94';

data hfias;
    set mydata.hfias;
run;

data hfias;
    set hfias;

    if AEZ = "high" and garden = 0 then AEZgarden = "high0";
    if AEZ = "high" and garden = 1 then AEZgarden = "high1";

    if AEZ = "med"  and garden = 0 then AEZgarden = "med0";
    if AEZ = "med"  and garden = 1 then AEZgarden = "med1";

    if AEZ = "low"  and garden = 0 then AEZgarden = "low0";
    if AEZ = "low"  and garden = 1 then AEZgarden = "low1";
run;

proc glm data=hfias;
    class AEZ garden;
    model foodsec = AEZ garden AEZ*garden;
    title "Original ANOVA: AEZ, garden, and AEZ*garden Interaction";
run;
quit;

proc glm data=hfias;
    class AEZgarden;
    model foodsec = AEZgarden;
    lsmeans AEZgarden / stderr tdiff adjust=tukey lines;
    title "One-Way ANOVA: AEZgarden (Combined AEZ    garden Factor)
       ";
run;
quit;


```

```
43
44 libname mydata '/home/u64258127/sasuser.v94';
45
46 data hfias;
47     set mydata.hfias_cluster;
48 run;
49
50 proc freq data=hfias;
51 run;
52
53 proc sort data=hfias;
54     by village;
55 run;
56
57 symbol1 v=dot i=none;
58 proc gplot data=hfias;
59     plot hfias*village = intervention;
60 run;
61 quit;
62
63 proc mixed data=hfias cl;
64     class village intervention;
65     model hfias = intervention / ddfm=kr;
66     random intercept / subject=village vcorr=1;
67     lsmeans intervention / cl;
68 run;
69 quit;
70
71 proc mixed data=hfias cl;
72     class intervention;
73     model hfias = intervention / ddfm=kr;
74     lsmeans intervention / cl;
75 run;
76 quit;
77
78 ************************************************************************
79
80 libname mydata '/home/u64258127/sasuser.v94';
81
82 data backpain;
83     set mydata.backpain;
84 run;
85
86 /* Check dataset structure */
87 proc contents data=backpain; run;
88 proc means data=backpain; var Y; class treatment period; run;
89
```

```
/* Paired crossover analysis using PROC MIXED */
proc mixed data=backpain;
    class person treatment period;
    model Y = treatment period / ddfm=kr;
    random person;
    lsmeans treatment / cl;
run;
```

Listing 1: SAS Code for Homework 8

Original ANOVA: AE2_garden, and AE2*garden Interaction

The GLM Procedure

Class Level Information		
Class	Levels	Values
AE2	2	1 2
garden	2	1 2

Number of Observations Read	400
Number of Observations Used	400

Original ANOVA: AE2_garden, and AE2*garden Interaction

The GLM Procedure

Dependent Variable: hoobes

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	836.410220	278.803407	16.22	<.0001
Error	396	6806.589780	17.187348		
Corrected Total	399	7643.000000			

R-Square	Coeff Var	Root MSE	hoobes Mean
0.109441	34.63346	4.145463	11.97000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
AE2	1	774.760000	774.760000	45.08	<.0001
garden	1	59.132580	59.132580	3.44	0.0644
AE2*garden	1	2.517640	2.517640	0.15	0.7000

Source	DF	Type III SS	Mean Square	F Value	Pr > F
AE2	1	774.760000	774.760000	45.08	<.0001
garden	1	59.132580	59.132580	3.44	0.0644
AE2*garden	1	2.517640	2.517640	0.15	0.7000

Interaction Plot for hoobes:

One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

The GLM Procedure

Class Level Information	
Class	Levels Values
AE2garden	4 hgh1 hgh2 med1 med2

Number of Observations Read	400
Number of Observations Used	400

One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

The GLM Procedure

Dependent Variable: hoobes

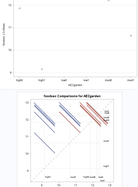
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	836.410220	278.803407	16.22	<.0001
Error	396	6806.589780	17.187348		
Corrected Total	399	7643.000000			

R-Square	Coeff Var	Root MSE	hoobes Mean
0.109441	34.63346	4.145463	11.97000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
AE2garden	3	836.410220	278.803407	16.22	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
AE2garden	3	836.410220	278.803407	16.22	<.0001

Distribution of hoobes



One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

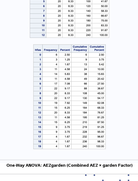
The GLM Procedure
Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer

AE2garden	hoobes LSMEAN	LSMEAN Number
hgh1	11.220000	1
hgh2	12.720000	2
med1	11.220000	3
med2	12.720000	4

Least Squares Means for Effect AE2garden
t for H0: LSMean(i)=LSMean(j) / Pr > |t|

Dependent Variable: hoobes				
i/j	1	2	3	4
1	-3.8647	-7.5000	0.0000	-7.5000
	0.0007	<.0001	1.0000	<.0001
2	3.8647	-3.6353	3.8647	-3.6353
	0.0007	0.0017	0.0007	0.0017
3	0.0000	7.5000	-3.8647	7.5000
	1.0000	<.0001	0.0007	<.0001
4	7.5000	3.6353	7.5000	-3.8647
	<.0001	0.0017	<.0001	0.0007

LS-Means for AE2garden



hoobes Comparisons for AE2garden

hoobes Tukey-Kramer Grouping for LS-Means of AE2garden (Alpha = 0.05)

LS-Means	Cldata
hgh2	12.7200
med2	12.7200
hgh1	11.2200
med1	11.2200

One-Way ANOVA: AE2garden (Combined AE2 + garden Factor)

The FREQ Procedure

nHap	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	1	0.25	1	0.25
2	0	0.00	1	0.25
3	0	0.00	1	0.25
4	0	0.00	1	0.25

Design and Analysis of Randomized Trials in Public Health

Homework 11

1. **Refer to our class example, "foodsecurity.sas" from Topic 9 (RCBDs).** Use the following code to recode AEZ and garden into a single variable, AEZgarden:

```
data hfias; set hfias;
       if AEZ="high" and garden=0 then AEZgarden="high0";
       if AEZ="high" and garden=1 then AEZgarden="high1";
       if AEZ="med" and garden=0 then AEZgarden="med0";
       if AEZ="med" and garden=1 then AEZgarden="med1";
       if AEZ="low" and garden=0 then AEZgarden="low0";
       if AEZ="low" and garden=1 then AEZgarden="low1";
run;
```

Reanalyze these data as a one-way ANOVA using AEZgarden, then compare the results to our earlier output. Use a Tukey adjustment when you do pairwise comparisons of means; suggested code:

```
lsmeans AEZgarden / stderr tdiff adjust=tukey lines;
```

a) How do the two ANOVA tables compare? (1 point)

The two ANOVA tables are identical:

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	839.243233	167.848647	18.22	<.0001
Error	594	5471.715100	9.211642		
Corrected Total	599	6310.958333			

b) In our class example, we had the following output from the "slice" option. What is the null hypothesis (in words) for the first statistical test reported below (where AEZ = "high")? (1 point)

The GLM Procedure
Least Squares Means

AEZ*garden Effect Sliced by AEZ for foodsec

AEZ	DF	Sum of Squares	Mean Square	F Value	Pr > F
high	1	364.261055	364.261055	39.54	<.0001
low	1	0.613272	0.613272	0.07	0.7965
med	1	124.595572	124.595572	13.53	0.0003

We are testing the hypothesis that average food insecurity among households who receive the gardening intervention is the same as average food insecurity among households who do not receive the gardening intervention *in the high-altitude agroecological zone*.

c) In your new SAS output, show the output that corresponds to the statistical test in part (b). What is the test statistic being used in your new output? How does it relate to the test statistic in part (b)? (1 point)

Least Squares Means for Effect AEZgarden
t for H0: LSMean(i)=LSMean(j) / Pr > |t|
Dependent Variable: foodsec

i/j	1	2	3	4	5	6
1		6.288367	-1.01201	-1.21899	-0.84536	2.809221
		<.0001	0.9139	0.8276	0.9588	0.0574
2	-6.28837		-7.50257	-7.42079	-7.17942	-3.50158
	<.0001		<.0001	<.0001	<.0001	0.0066
3	1.012009	7.502566		-0.25802	0.147568	3.91437
	0.9139	<.0001		0.9998	1.0000	0.0014
4	1.21899	7.420794	0.258023		0.393367	3.993984
	0.8276	<.0001	0.9998		0.9988	0.0010
5	0.845362	7.179416	-0.14757	-0.39337		3.677755
	0.9588	<.0001	1.0000	0.9988		0.0035
6	-2.80922	3.501581	-3.91437	-3.99398	-3.67775	
	0.0574	0.0066	0.0014	0.0010	0.0035	

This is a t-statistic for the comparison of mean 1 (high AEZ, no garden intervention) and mean 2 (high AEZ, garden intervention). Note the p-value is the same, and if we square the t-statistic here, we get the F statistic in part (b).

2. Recall the cluster RCT example we discussed in class, "hfias cluster randomization.sas" (Topic 11).

a) We analyzed these data using Proc MIXED. Reanalyze the data, still using Proc MIXED, but ignore the clustering and treat the study as a completely randomized design. Make sure to keep the line "`lsmeans intervention / cl;`". For each model, what is your estimate for σ^2? Which model as a higher estimate, and why? (3 points)

```
title1 'with clustering';
proc mixed data=hfias cl;
       class village intervention;
       model hfias = intervention / ddfm=kr;
       random intercept / subject=village;
       lsmeans intervention / cl;
run;
title1;

title1 'ignoring clustering';
proc mixed data=hfias cl;
       class village intervention;
```

```
        model hfias = intervention / ddfm=kr;
        *random intercept / subject=village;
        lsmeans intervention / cl;
run;
title1;
```

When accounting for clustering: 9.1826
When ignoring clustering: 12.6445

When we ignore clustering, the variability that is actually due to differences among villages is no longer accounted for in our model and therefore gets lumped into the error variance, so the error variance increases when we ignore the clustering.

b) Compare the least squares means for the intervention and control groups in the two models. Does ignoring the clustering change your estimated average food insecurity in each group? Does ignoring the clustering change the standard errors of those estimates? In what way? (2 points)

The estimates don't change, but their standard errors do – the standard errors are (erroneously) much smaller when we ignore the clustering. If we do a comparison of means with smaller standard errors, we're more likely to reject the null hypothesis that the two means are equal. In other words, we're more at risk of making a type I error when we ignore clustering.

c) If you incorrectly calculate the standard error of an estimate to be smaller than it actually is, you are going to erroneously think you have more precise estimates than you actually do. What are the (statistical) risks of thinking you have more precision than you actually do? (2 points)

As described above, we're more at risk of making a type I error (a false positive).

d) Compare the 95% confidence intervals for the group means in the two models. Which model gives you wider confidence intervals? Why? (2 points)

When we correctly account for clustering, we have wider confidence intervals for the group means. This is because those confidence intervals (based on the larger SE discussed in parts b and c) are accounting for both variance among villages (σ_τ^2) and variance within villages (σ^2).

3. You have developed an intervention intended to increase the amount of time patients with chronic back pain do a particular type of exercise, and you evaluate your intervention compared to standard of care physical therapy in a crossover study. In the dataset "backpain", "person" is the participant ID, and you have 20 participants. Every participant receives both the intervention (treatment="A") and the standard of care (treatment="B") in randomized order. "Period" indicates the time period in which your outcome, Y=number of minutes per week spent doing the recommended exercise, is assessed. What effect does your intervention have on the amount of time patients do the recommended exercise? Detail your steps, reasoning, and interpretation of results, as well as things you explore and check along the way. Include the SAS code you used. (10 points)

Starting with the data import, and for each step below, I'm always checking the log! Also from the beginning, I'm deciding to use a 0.05 significance level.

```
libname hw 'C:\Users\gunaratn\Box\Randomized Trials\homework\HW11';
data hw; set hw.backpain; run;
```

I open the data and see there are 40 observations and 4 variables, as expected. I can see that there are no missing data, and the values for Y are all integers.

Because the data set is so small, I can just sort by person and period, and in the same time it would take me to write code, I can visually observe that each person has both time periods and both treatments, and people differ in the order they do the treatments:

```
proc sort data=hw; by person period; run;
```

Proc FREQ of all the variables confirms this – 2 observations/person, 20 observations/period, 20 observations/treatment, and Y has reasonable values, ranging from 29 to 59 minutes, with no zeros, negative numbers, or very high-looking numbers:

```
proc freq data=hw; run;
```

To get to know Y a bit better, I use proc UNIVARIATE and check n, mean, median, SD, min, max, the quantiles, extreme observations, a histogram …

```
proc univariate plot data=hw;
        var Y;
        histogram Y;
run;
```

My hope is that the intervention increases the time spent doing the recommended exercise on average. I know people's behavior can change over time too, in general or in response to what they did previously. I try proc MEANS:

```
proc sort data=hw; by period treatment; run;
proc means data=hw; by period treatment;
        var Y;
        output out=out1 mean=;
run;
```

10 observations per treatment and period combination, as expected. Standard deviations are close-ish, given the small number of observations in each combination. In period 1, treatment B has a higher mean than treatment A, but in period 2, the two means seem pretty similar. Sounds like an interaction between treatment and period, but I'll have to do some statistical testing to be sure. Treatment B is the standard of care, so if anything, in period 1, the intervention actually does the same or worse than the standard of care. In period 2, the intervention doesn't seem so different from the standard of care.

period=1 treatment=A

Analysis Variable : Y

N	Mean	Std Dev	Minimum	Maximum
10	38.7000000	7.6165026	29.0000000	55.0000000

period=1 treatment=B

Analysis Variable : Y

N	Mean	Std Dev	Minimum	Maximum
10	48.1000000	7.6659420	35.0000000	59.0000000

period=2 treatment=A

Analysis Variable : Y

N	Mean	Std Dev	Minimum	Maximum
10	49.9000000	5.8395205	38.0000000	57.0000000

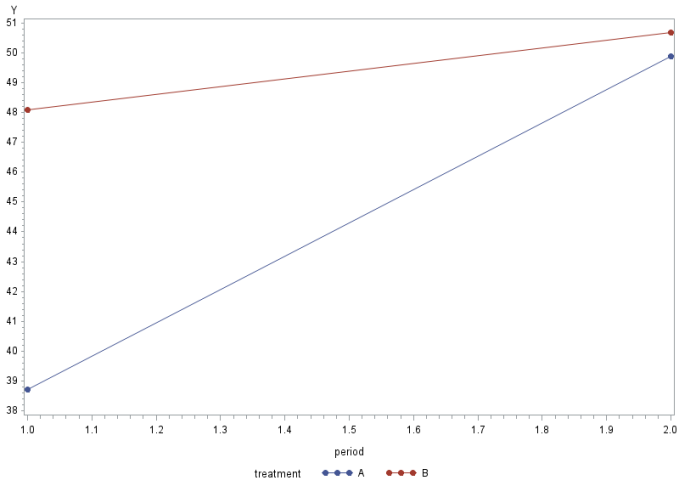
period=2 treatment=B

Analysis Variable : Y

N	Mean	Std Dev	Minimum	Maximum
10	50.7000000	6.7007462	38.0000000	58.0000000

I graphed these means too, just to see it better:

```
symbol1 v=dot i=join;
symbol2 v=dot i=join;
proc gplot data=out1;
        plot Y*period=treatment;
run;
quit;
```



This does not bode well for the intervention, but now we can formally analyze the data:

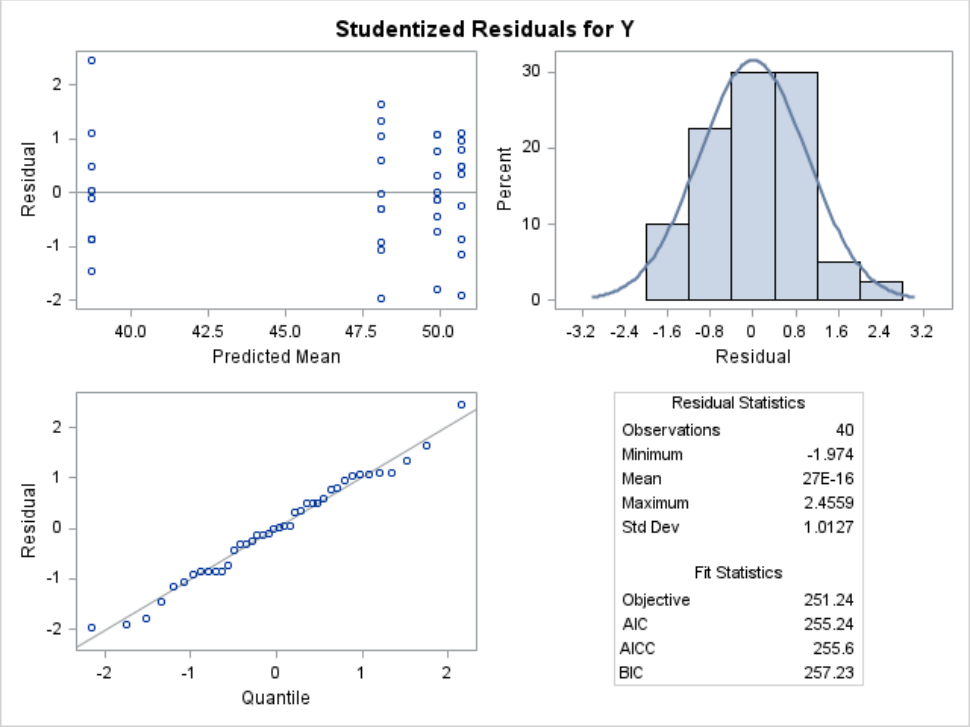
```
proc mixed covtest data=hw plots=all;
        class person period treatment;
        model Y = treatment period treatment*period / ddfm=kr;
        repeated period / subject=person type=cs r=1;
run;
```

The model converges, and the log is happy. Because there are only 2 observations/person, there are effectively fewer options for the variance-covariance matrix. I'm going to assume everyone has the same variance-covariance matrix (no heterogeneity among the participants that affects their covariance structure), and I have no reason to think the variance changes depending on the period, so I'm going to choose a variance-covariance matrix that has the same number along the diagonal. And variance-covariance matrices are symmetric around the diagonal (top right number = bottom left number), so the matrix is going to look like this:

$$\begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

where *a* and *b* are some numbers. Typical options for the variance-covariance structure would be compound symmetry (exchangeable) or AR(1) (first order autoregressive), but if we fit either, we'll end up with the same answer (try it and see!). So I used type=cs because I had to pick something (if I leave out "type=", SAS will assume there is no correlation between the 2 observations from the same person).

I check everything in my proc MIXED output, to make sure it matches my expectations (model info, class level info, etc. etc.). I had asked proc MIXED for "plots=all", so I check those out. Residuals look to have constant variance and approximate normality, and there is no evidence of any notable outliers:



SAS also plots residuals vs. treatment, period, treatment*period, and person! The sample sizes are small, and I see no particular patterns here. The last plot, residuals for each person, is useful – I don't see any person that really stands out or seems different from other participants.

Looking at my type 3 tests and sticking to my pre-specified significance level of 0.05, the interaction is not significant, but the main effects each are:

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
treatment	1	18	5.94	0.0255
period	1	18	10.86	0.0040
period*treatment	1	18	3.42	0.0809

It's possible that the interaction is truly significant and, with the small sample size, I didn't have the power to detect it. However, since I pre-specified a 0.05 significance level, I'm going to accept it as not significant and explore the main effects:

```
proc mixed covtest data=hw plots=all;
        class person period treatment;
        model Y = treatment period treatment*period / ddfm=kr;
        repeated period / subject=person type=cs r=1;
        lsmeans treatment / diff cl;
        lsmeans period / diff cl;
run;
```

Least Squares Means

Effect	treatment	period	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
treatment	A		44.3000	1.5644	35.6	28.32	<.0001	0.05	41.1261	47.4739
treatment	B		49.4000	1.5644	35.6	31.58	<.0001	0.05	46.2261	52.5739
period		1	43.4000	1.5644	35.6	27.74	<.0001	0.05	40.2261	46.5739
period		2	50.3000	1.5644	35.6	32.15	<.0001	0.05	47.1261	53.4739

Differences of Least Squares Means

Effect	treatment	period	_treatment	_period	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
treatment	A		B		-5.1000	2.0934	18	-2.44	0.0255	0.05	-9.4980	-0.7020
period		1		2	-6.9000	2.0934	18	-3.30	0.0040	0.05	-11.2980	-2.5020

Our main interest is the effect of treatment. We find that patients who receive our intervention spend less time per week on this recommended exercise than those who receive the standard of care – on average, about 5 minutes/week less (95% CI: 0.7-9.5 minutes less per week). That may or may not be a meaningful difference (is 5 minutes/week a lot? A clinician can tell you), but the point is that our intervention reduced, not increased, the time spent on the exercise – definitely not an intervention to promote or continue.

It's beyond the scope of this class, but we can also explore whether doing treatment A first, then B has a residual effect and leads to different results than doing B first, then A.